OPEN SOURCE · LIVE MARITIME INTELLIGENCE

# Voyance II

Know where your ETA model fails — before the storm does.

| 4.49 h | 83% | 8,020 | 25+ |
| --- | --- | --- | --- |
| TEST MAE · GPU MODEL | P10–P90 COVERAGE | REAL VOYAGES MODELED | OPEN DATA SOURCES |

github.com/devansh0703/voyance2 · Apache-2.0 · every number reproducible from pipeline artifacts

THE PROBLEM · IDEALAB IDEA #586938

# Operators judge AI models from vendor claims.

A fleet team trusts a vendor's arrival estimate but cannot see where it fails in bad weather or certain waters.

Windward, PortXchange, WHIZTEC sell closed operational intelligence. Nobody ships an open, served, weather-conditioned model with honest error decomposition.

**$100M**

WEDGE MARKET (MODEL ESTIMATE)

**$4.3B**

WIDER MARITIME AI MARKET

Source: idealab.ai/idea-details/586938 market analysis (Grand View Research 2024 proxy)

THE PRODUCT

# A live maritime intelligence terminal, open end to end

**01 INGEST**

NOAA MarineCadastre AIS
+ UN/LOCODE ports

→ **02 VOYAGEIZE**

port-buffer segmentation
plausibility filters

→ **03 WEATHERIZE**

ERA5 wind · waves · swell
hourly grid join

→ **04 SERVE**

GPU quantile ETA
FastAPI /v1 /v2

→ **05 WATCH**

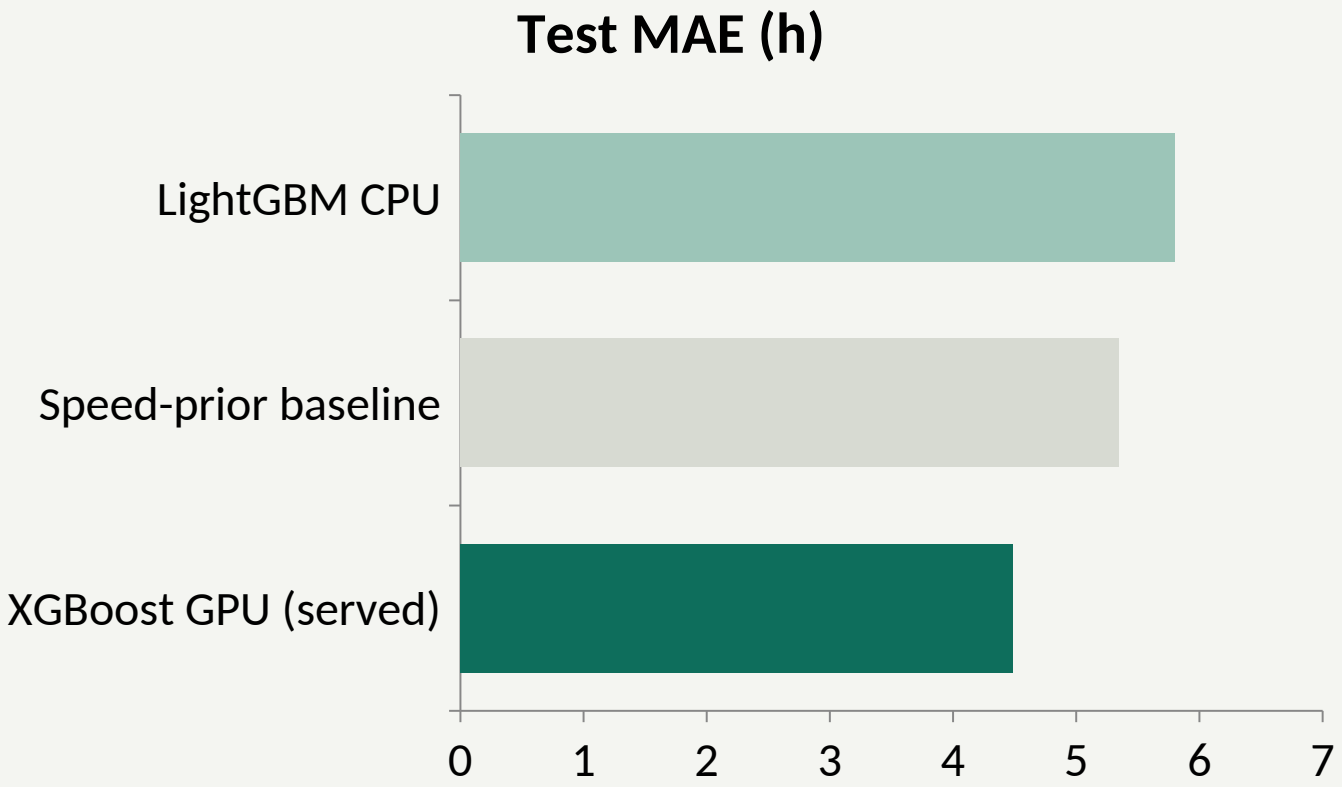
drift PSI/KS
Evidently + alerts

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Every layer is open data and open code: NOAA/CC0 · Open-Meteo/CC BY 4.0 · OSM/ODbL · Digitraffic/CC BY 4.0 · EMODnet · USGS · OBIS/GBIF. Zero signups, zero fabricated numbers.

THE MODEL · TRAINED ON THE OWNER'S RTX 4060 (CUDA)

# Beats the speed prior by 16% — and stays calibrated




**4.49 h**

TEST MAE — JULY 2024 HOLDOUT

· Quantile P10/P50/P90, conformal-calibrated (CQR, Romano 2019)
· 83% empirical coverage vs 80% nominal
· Physics-informed ratio features — trees can't divide
· 18,040 train / 12,924 test rows, voyage-disjoint split

Source: reports/eval.json (generated by pipeline/train.py — not hand-entered)

THE PRODUCT IS THE PREDICTION, LAID OUT

# Not a point estimate — a full voyage surface

· Actual track + remaining leg on real chart geography
· Drift cone sized by the P10–P90 lateral spread
· Historical analog voyages drawn dashed — same port, same ship type
· ETA quantile cards with distance-to-go and fix count
· One GeoJSON endpoint: /v2/predict/route/{voyage}

**P10/P50/P90**

CONFORMAL ETA BAND

**±2.73 h**

CQR INTERVAL ADJUSTMENT

Live at http://127.0.0.1:8010/predict · GeoJSON served by XGBoost quantile models

LIVE OPERATIONS · NO SIGNUP FEEDS

# Real-time fleet picture with collision watch

| | |
|---|---|
| **LIVE AIS** | 300 vessels · Digitraffic Finland · 20 s refresh |
| **COLLISION** | DCPA/TCPA solver · Goodwin-Stanton ellipses · fuzzy CRI |
| **TRACKING** | EKF + UKF filters · Douglas-Peucker · Fréchet residuals |
| **OVERLAYS** | MPA polygons · quakes · radar loop · SAR chips · harbors |
| **QUALITY** | IQR/z-score audit · upstream health · event ticker |

Encounter view draws safety-domain ellipses + CPA extension lines on the chart — /live

EXPLAINABLE BY CONSTRUCTION

# An AI analyst that cites real voyages

“Why will this route drift tomorrow for this ship type?”

The answer is generated only from a context packet: live forecast + waves at the vessel, the model's tested MAE and top features, and the actual historical analog voyages — each named, each replayable on the map.

Provider chain: OPENCODE_API_KEY first, chat-verified Groq/OpenRouter fallback — every response states who served it.

PARSED OUTPUT

```
## Headline

## Why drift is expected

## Reference backing

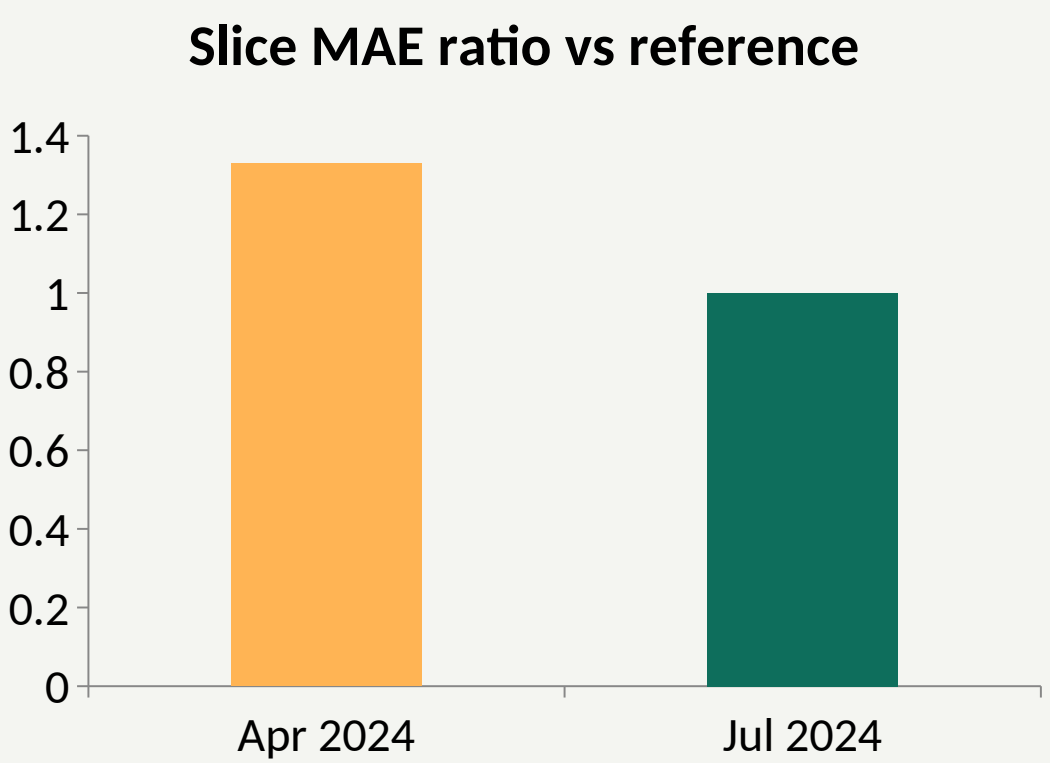
## What to watch
```

```
references = real voyage_ids
replayable in one click
```

POST /v2/ai/explain · answers forbidden from inventing numbers outside the context packet

PRODUCTION HONESTY

# Drift monitoring with alerts, not vibes




- Evidently DataDrift per feature: PSI + KS
- Alert rule: PSI > 0.2 or MAE ratio > 1.25
- April slice drifted on distance/SOG mix — caught, explained, surfaced as a chip on the dashboard
- Machine-readable alerts at /v1/drift

Source: reports/drift_summary.json + Evidently HTML reports in reports/drift/

POSITIONING

# The open slot in the landscape

## CLOSED VENDORS

Windward · PortXchange
WHIZTEC · Broadpath
Kairon

Operational visibility,
not independent error audit

## OPEN BENCHMARKS

WUT-TrajData

Dataset + baselines,
not a served product

## VOYANCE II

**Apache-2.0 · self-host**

Served quantile model +
scorecard + AI analyst

Landscape from idealab #586938 competitor analysis · full matrix on the landing page

RUN IT YOURSELF

# git clone · make data · make train · make api

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Or just: ./start.sh — one script boots the API, dashboards, live feeds and the AI analyst on your machine. GPU training included.

github.com/devansh0703/voyance2 · Apache-2.0 · built on open data only